

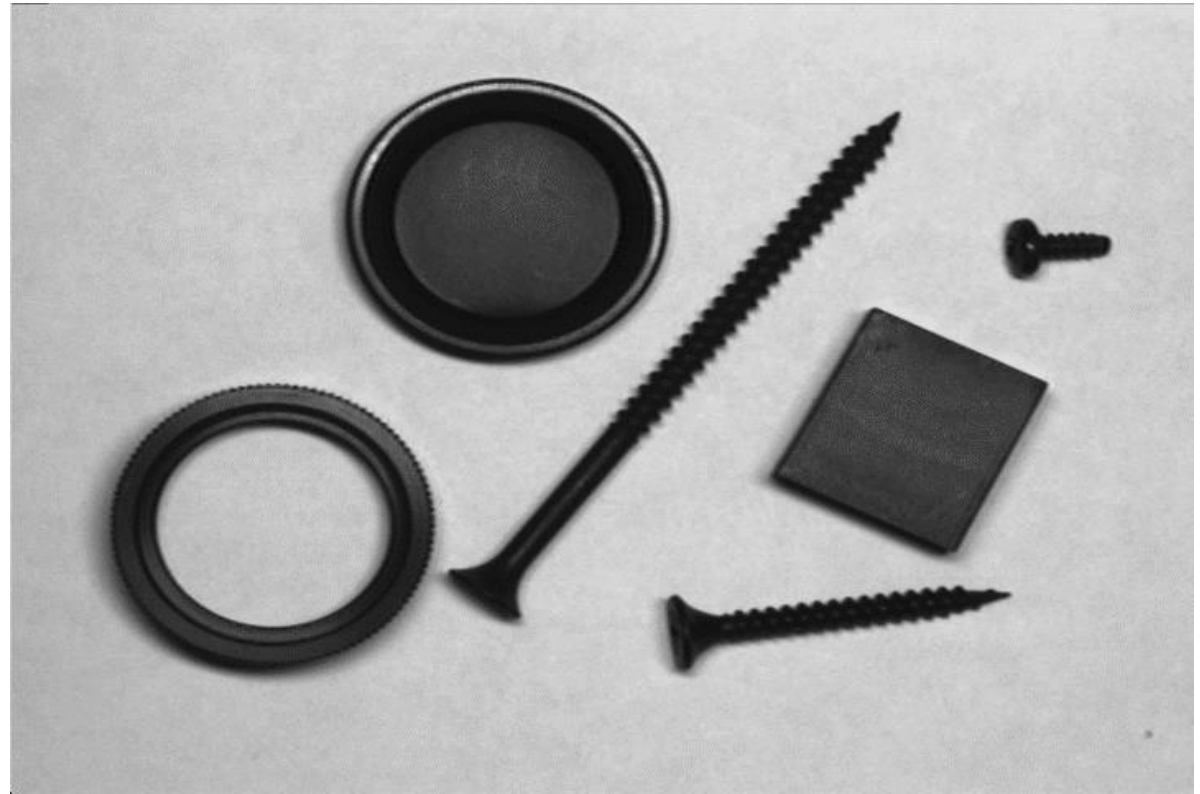
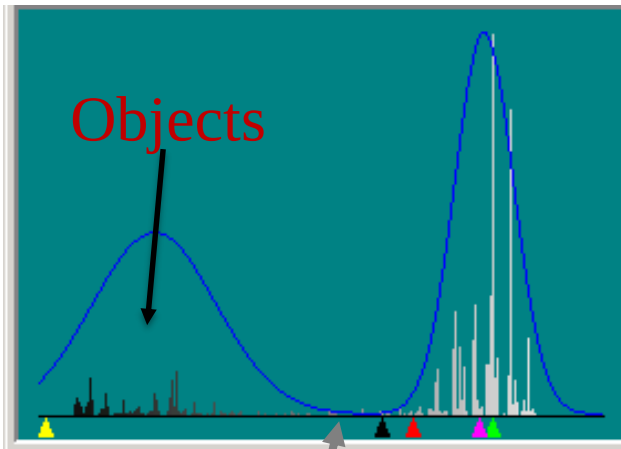
Lecture 9:

MORPHOLOGICAL IMAGE PROCESSING

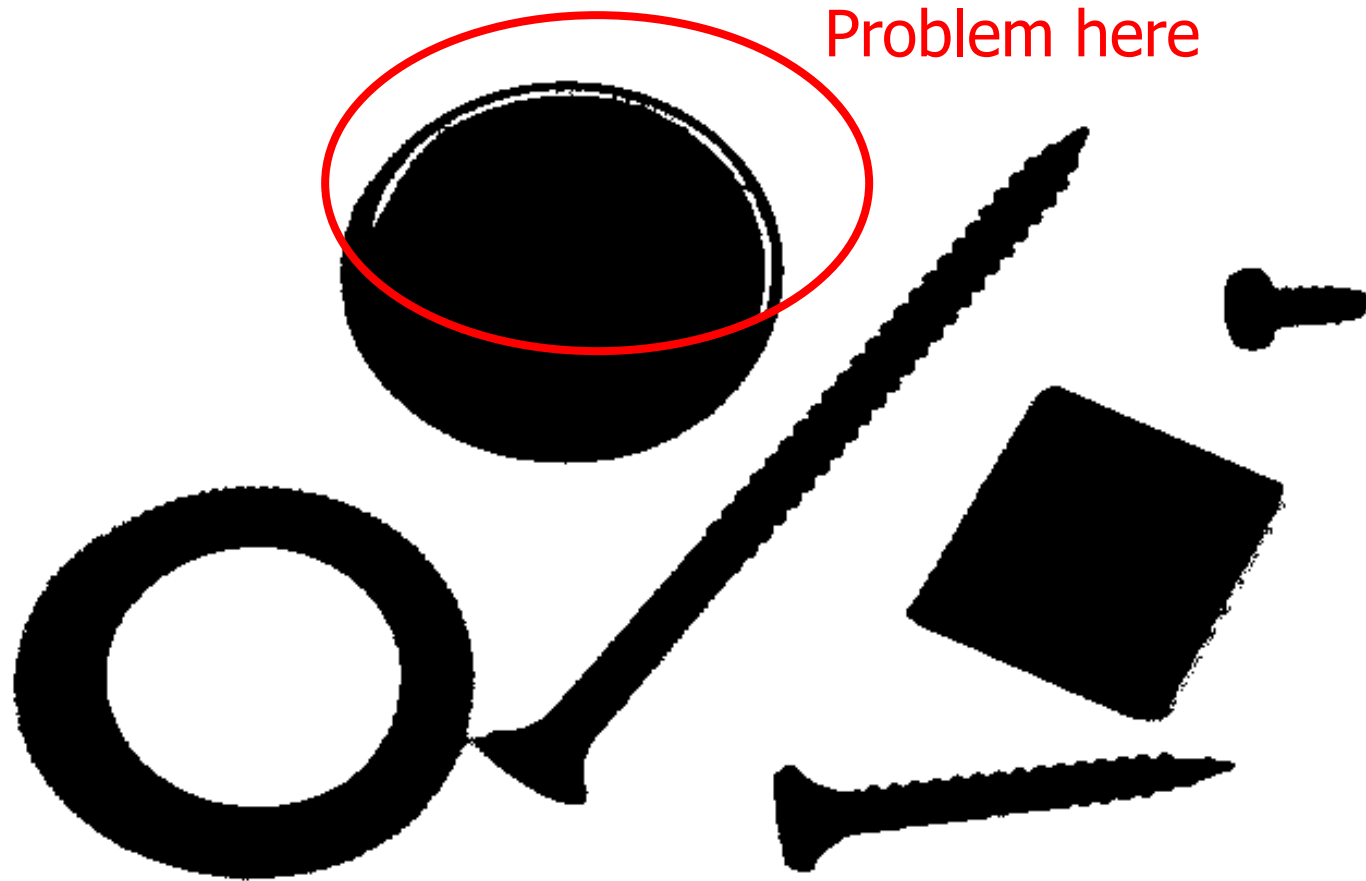
Nội dung

- Set-theoretic interpretation
- Binary dilation and erosion
- Opening, closing, morphological edge detectors

Gray Level Thresholding



Binary Image



Problem here

How do we fill "missing pixels"?

Original image

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



After dilation

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.

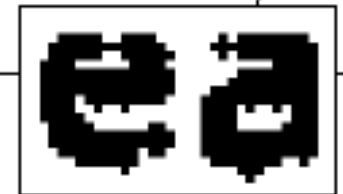


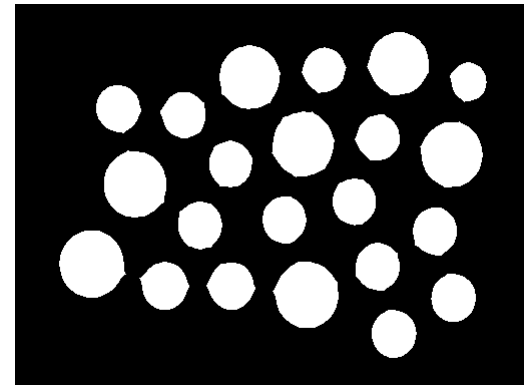
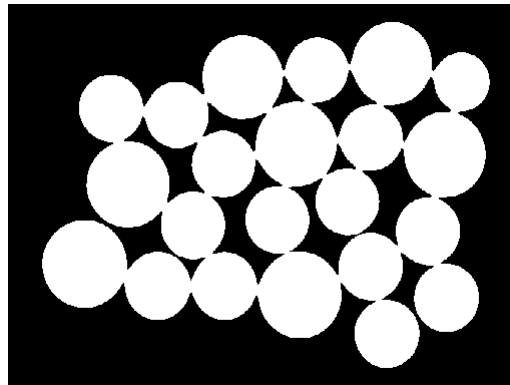
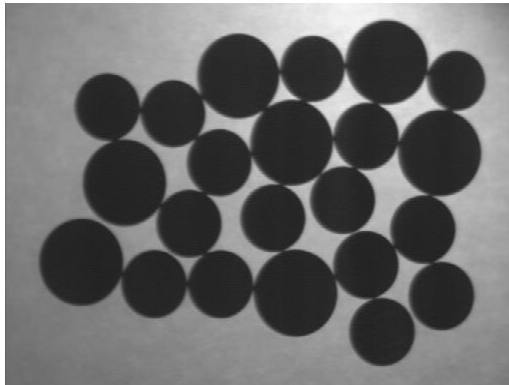


Image after segmentation



Image after segmentation and
morphological processing

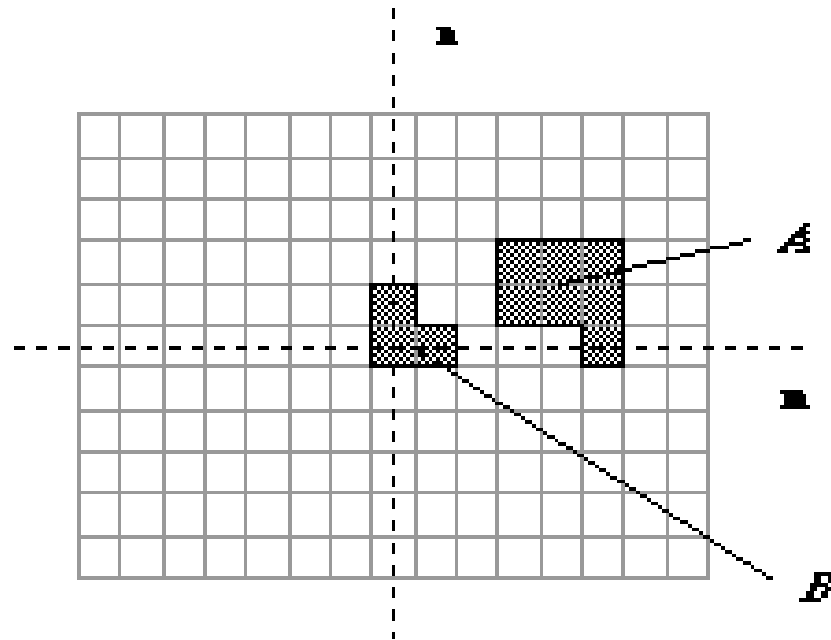
Counting Coins



Mathematical Morphology

- Ảnh là một hàm hai chiều, $f(x, y)$, của các biến tọa độ rời rạc (x, y) .
- Một định nghĩa thay thế có thể dựa trên khái niệm rằng ảnh bao gồm một tập hợp các tọa độ rời rạc.

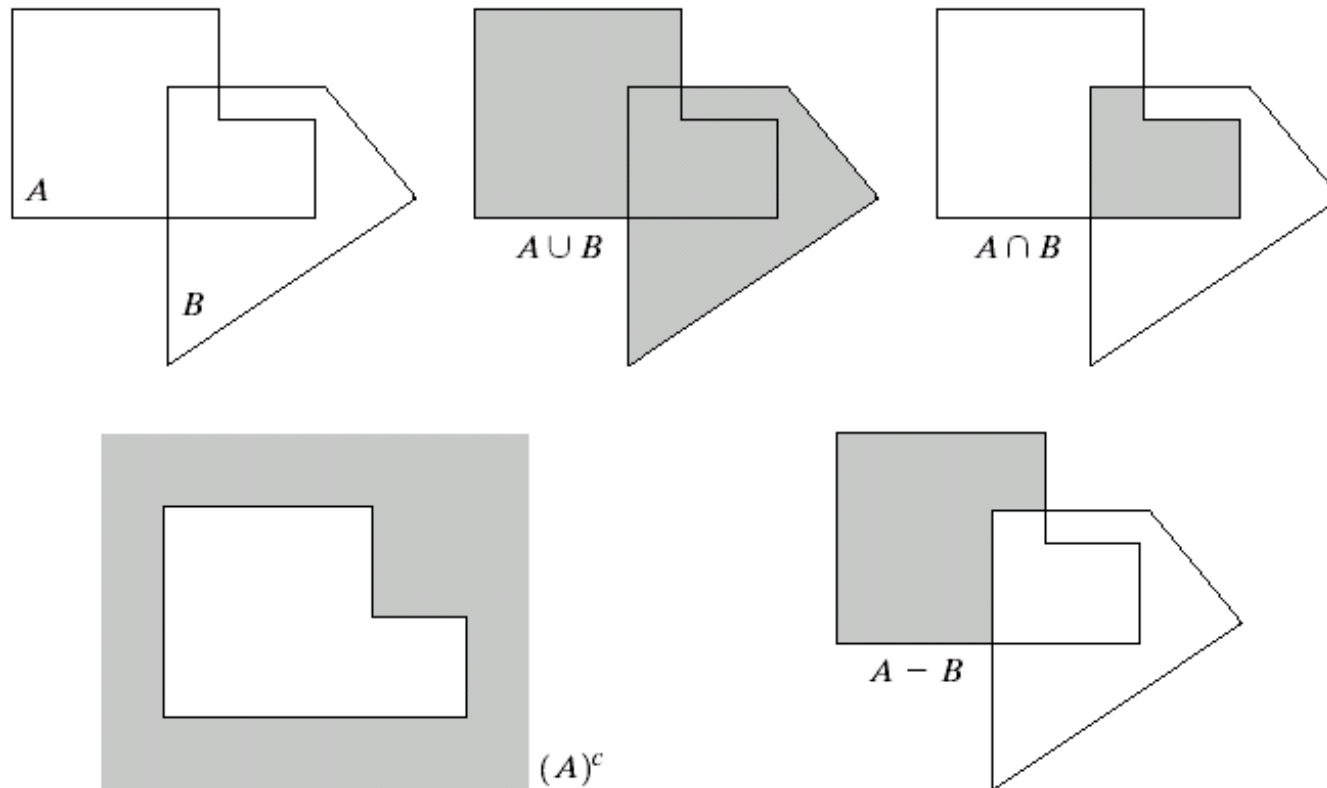
Morphology



A binary image containing two object sets A and B

- $B = \{(0,0), (0,1), (1,0)\}$
- $A = \{(5,0), (3,1), (4,1), (5,1), (3,2), (4,2), (5,2)\}$

Basic Set Theory



a	b	c
d	e	

FIGURE 9.1

(a) Two sets A and B . (b) The union of A and B . (c) The intersection of A and B . (d) The complement of A . (e) The difference between A and B .

Logic Operations

p	q	p AND q (also $p \cdot q$)	p OR q (also $p + q$)	NOT (p) (also $\bar{p}$)
0	0	0	0	1
0	1	0	1	1
1	0	0	1	0
1	1	1	1	0

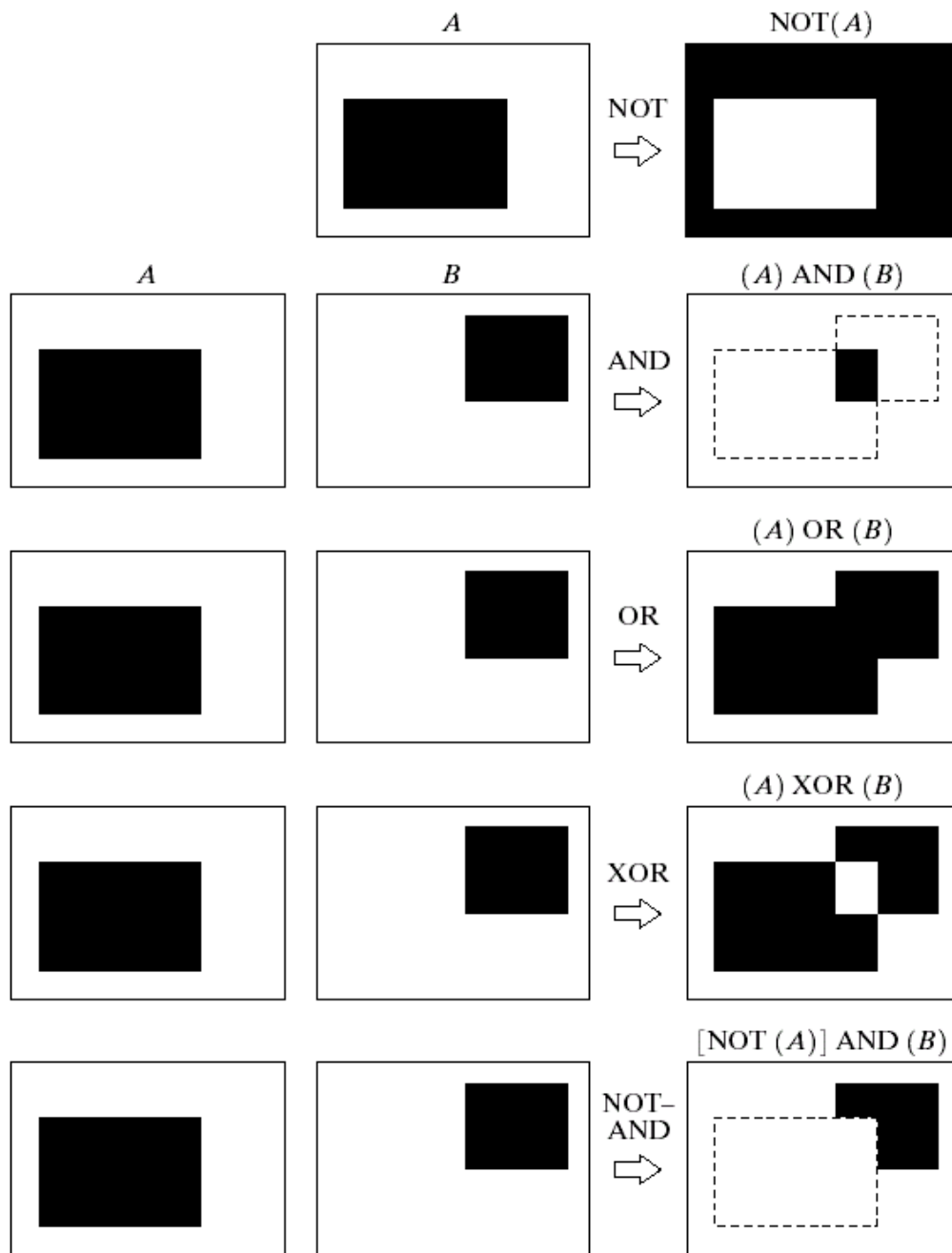


FIGURE 9.3 Some logic operations between binary images. Black represents binary 1s and white binary 0s in this example.

Some Basic Definitions

- Let A and B be sets with components $a=(a_1,a_2)$ and $b=(b_1,b_2)$, respectively.

- The *translation* of A by $x=(x_1,x_2)$ is

$$A + x = \{c \mid c = a + x, \text{ for } a \in A\}$$

- The *reflection* of A is

$$A^r = \{x \mid x = -a \text{ for } a \in A\}$$

- The *complement* of A is

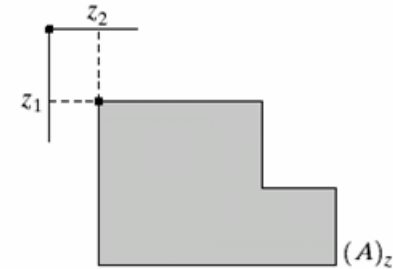
$$A^c = \{x \mid x \notin A\}$$

- The *union* of A and B is

$$A \cup B = \{x \mid x \in A \text{ or } x \in B\}$$

- The *intersection* of A and B is

$$A \cap B = \{x \mid x \in A \text{ and } x \in B\}$$



Morphological Image Processing

- Hình thái học toán học được áp dụng cho hình ảnh nhị phân (có thể mở rộng cho ảnh đa mức xám).
- Đây là một trong những kỹ thuật được áp dụng trong giai đoạn tiền xử lý.
- Hai phép toán thường dùng là phép giãn nở (Dilation) và phép co (Erosion). Từ hai phép toán cơ bản này phát triển thành một số phép toán như phép đóng (Closing) và phép mở (Opening).
- Dilation gọi là $D(i)$: giãn nở.
- Erosion gọi là $E(i)$: co.
- Một chu trình $E(i)$ - $D(i)$ gọi là phép mở (Opening).
- Một chu trình $D(i)$ - $E(i)$ gọi là phép đóng (Closing).

Ứng dụng

- Trích lọc biên ảnh (Boundary extraction).
- Tô đầy vùng (Region fill).
- Trích lọc các thành phần liên thông (Extracting connected components).
- Làm mỏng đối tượng trong ảnh (Thinning).
- Làm dày đối tượng trong ảnh (Thickening).
- Tìm xương đối tượng trong ảnh (Skeletons).
- Cắt tỉa đối tượng trong ảnh (Pruning).

Binary image processing

- Representation of individual pixels as 0 or 1
 - Object = 1 (white)
 - background = 0 (black)
- Processing by logical functions is fast and simple

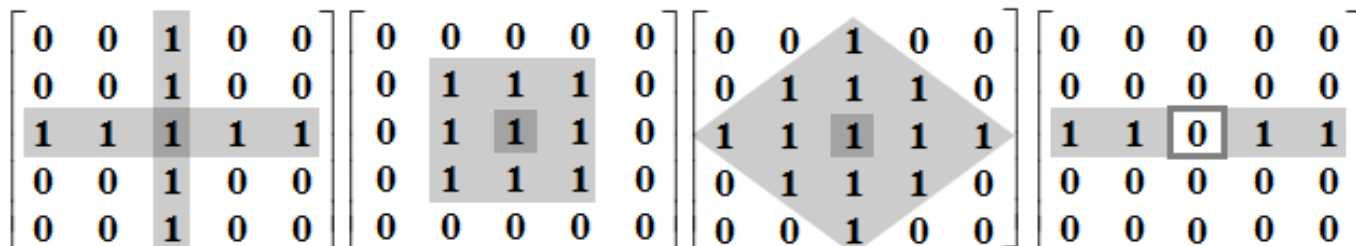
Structuring Element (SE)

- SE có thể thay đổi kích thước
- Giá trị của các phần tử là 0, 1

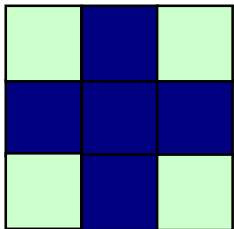
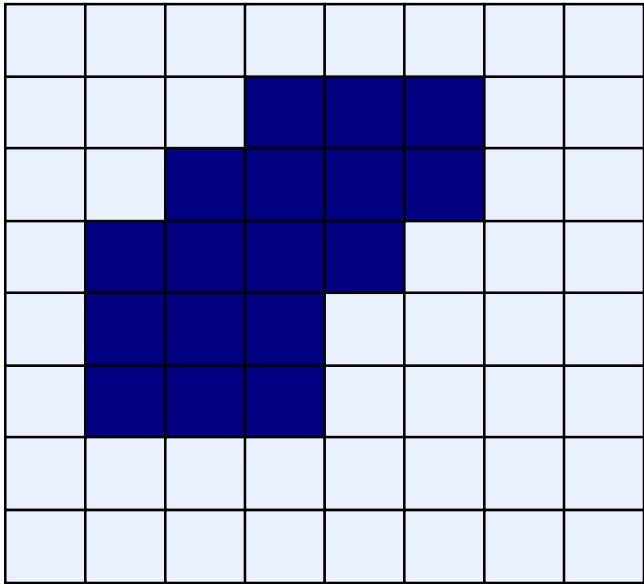
Examples of SE

Một số hình dáng của phần tử cấu trúc thường xuyên sử dụng trên ảnh nhị phân:

- Dạng đường theo chiều ngang và dọc.
- Dạng chữ thập.
- Dạng hình vuông.
- Dạng hình ellipse.
- ...



Fits & Hits



SE

- **FIT:** xảy ra khi tất cả các pixel trùng với các pixel của ảnh.
- **HIT:** xảy ra khi có bất kỳ pixel nào của SE trùng với pixel của ảnh.
- Các phép toán xử lý hình thái được dựa trên hai hiện tượng trên!

Fits & Hits

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0	0	0
0	0	1	1	1	1	1	0	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	1	1	1	1	0
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0

1	1	1
1	1	1
1	1	1

Structuring
Element 1

0	1	0
1	1	1
0	1	0

Structuring
Element 2

Fits & Hits

0	0	0	0	0	0	0	0	0	0	0	0
0	0	0	1	1	0	0	0	0	0	0	0
0	0	1	B	1	1	1	0	C	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	1	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	0	0	0	0
0	0	1	1	1	1	1	1	1	0	0	0
0	0	1	1	1	1	1	A	1	1	1	0
0	0	0	0	0	1	1	1	1	1	1	0
0	0	0	0	0	0	0	0	0	0	0	0

1	1	1
1	1	1
1	1	1

Structuring
Element 1

0	1	0
1	1	1
0	1	0

Structuring
Element 2

Các phép xử lý hình thái cơ bản

- Tương tự như các bộ lọc không gian.
- SE được chuyển qua tất cả các pixel của ảnh gốc để tạo ra ảnh mới.
- Giá trị của các pixel mới (tương ứng tại vị trí tâm của SE) phụ thuộc vào phép toán xử lý hình thái.
- Hai phép xử lý hình thái cơ bản là: **co (Erosion)** và **giãn (Dilation)**

Erosion – Phép co

- Phép co ảnh f với phần tử cấu trúc s , ký hiệu: $f \ominus s$
- Mô tả:

Giả sử SE s đang ở vị trí (x,y) , giá trị của pixel mới sau khi thực hiện phép toán có giá trị như sau:

$$g(x,y) = \begin{cases} 1 & \text{if } s \text{ fits } f \\ 0 & \text{otherwise} \end{cases}$$

Erosion – Phép co

Công thức:

$$\mathbf{A} \ominus \mathbf{B} = \{ \mathbf{c} \mid (\mathbf{B})_{\mathbf{c}} \subseteq \mathbf{A} \}$$

Trong đó:

- $\mathbf{A}$: Ma trận điểm ảnh của ảnh nhị phân.
- $\mathbf{B}$: Là phần tử cấu trúc.

Phép co ảnh sẽ cho ra một tập điểm ảnh $\mathbf{c}$ thuộc $\mathbf{A}$, nếu di chuyển phần tử cấu trúc $\mathbf{B}$ theo $\mathbf{c}$, thì $\mathbf{B}$ nằm trong đối tượng $\mathbf{A}$.

$\mathbf{E}(\mathbf{i})$ là một tập con của tập ảnh bị co $\mathbf{A}$.

Some Basic Definitions

- **Erosion – co/thu hẹp**

$$A \ominus B = \{x \mid (B + x) \subseteq A\}$$

- Erosion shrinks a region.

$$g[x,y] = AND\left[\hat{W}\left\{f[x,y]\right\}\right] := erode(f,W)$$



Original (701x781)



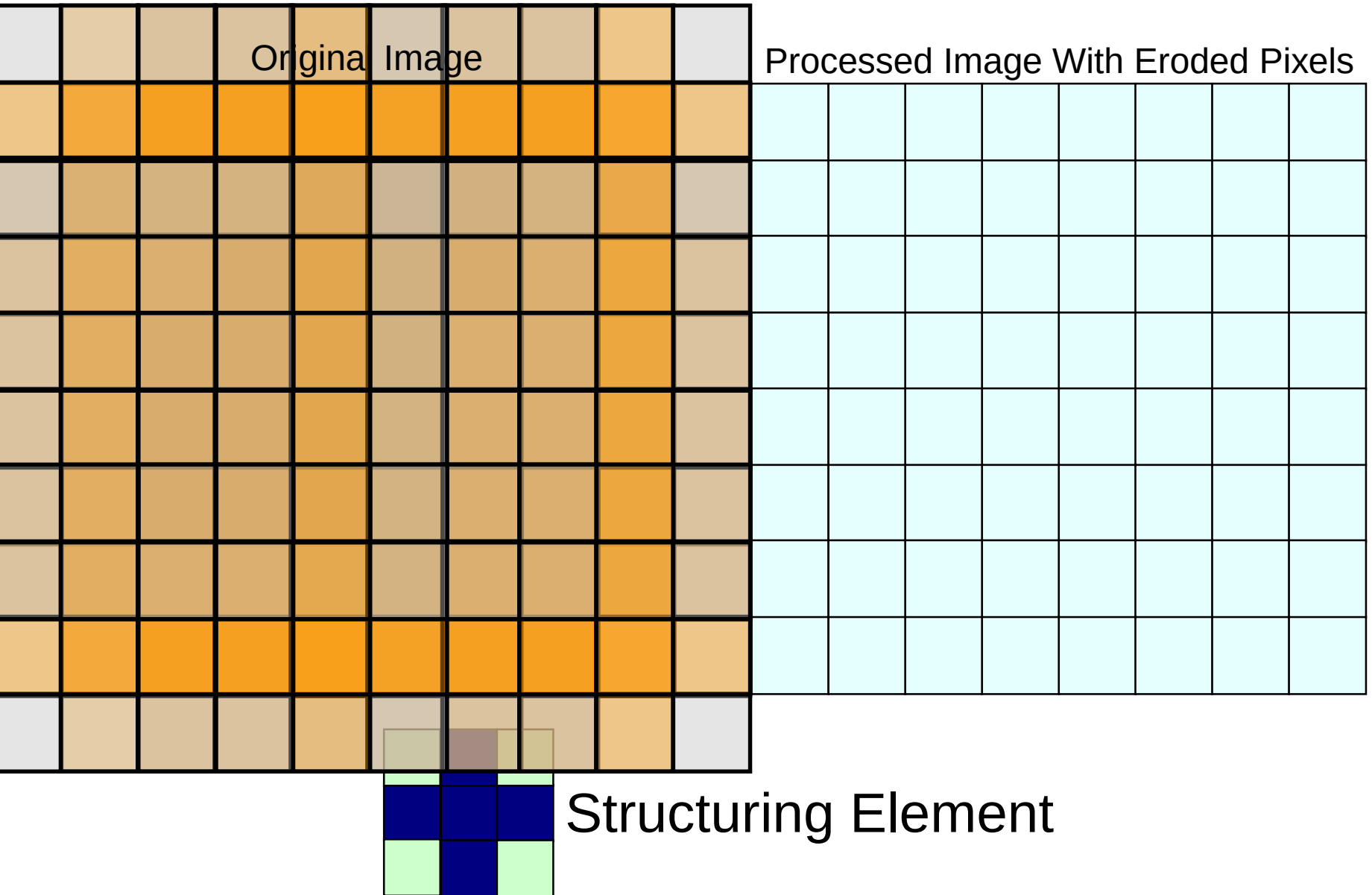
erosion with
3x3 structuring element



erosion with
7x7 structuring element

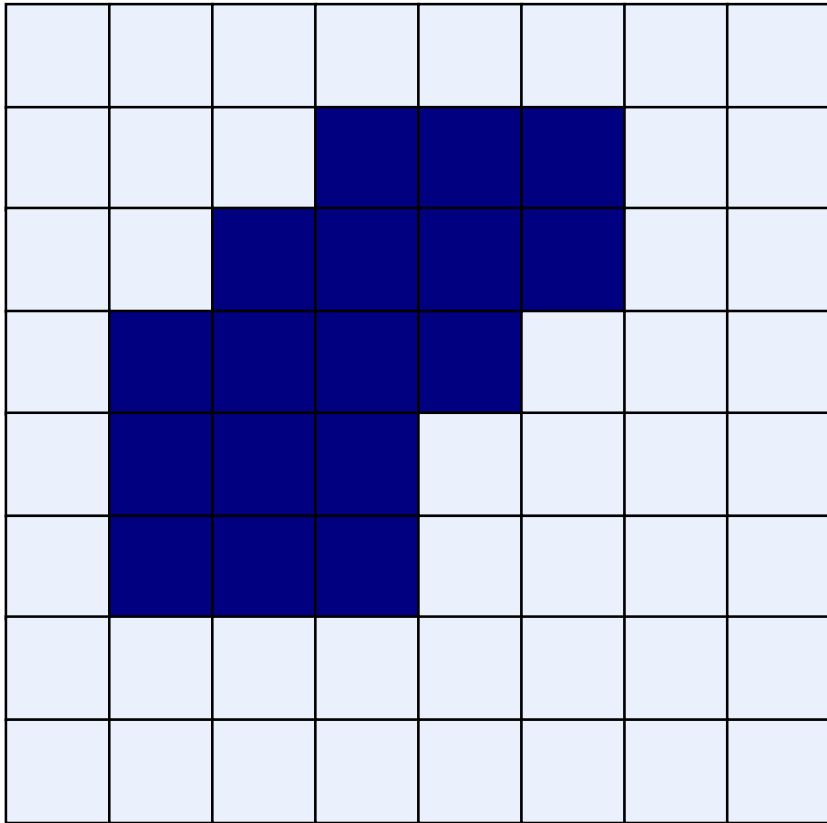
- Shrinks the size of 1-valued objects
- Smooths object boundaries
- Removes peninsulas, fingers, and small objects

Erosion Example

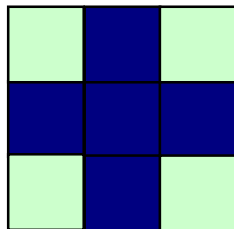
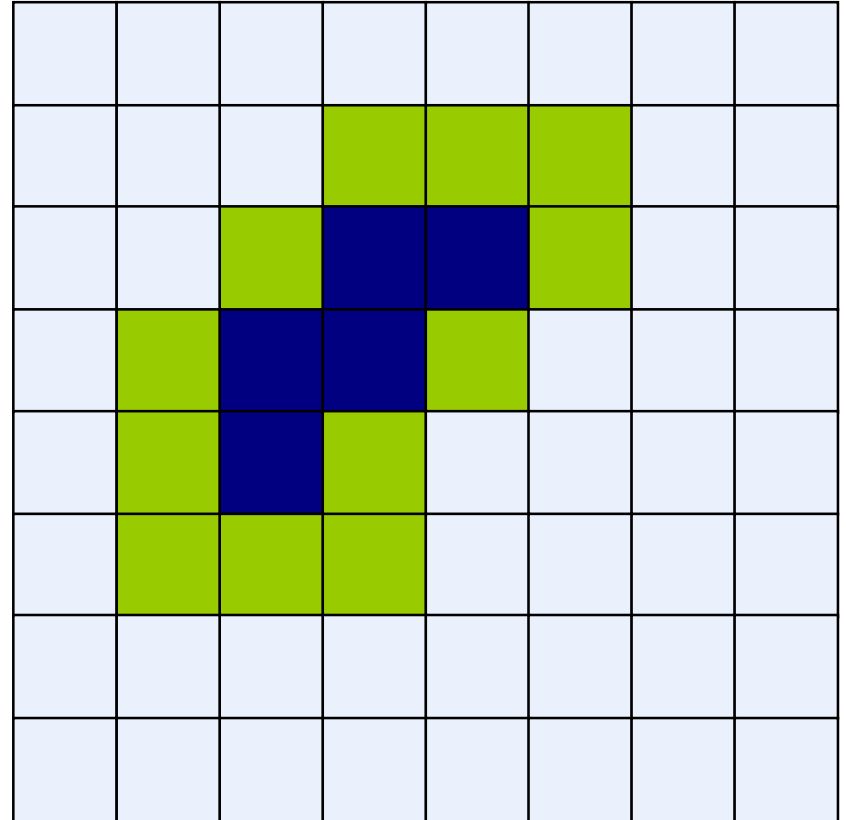


Erosion Example

Original Image

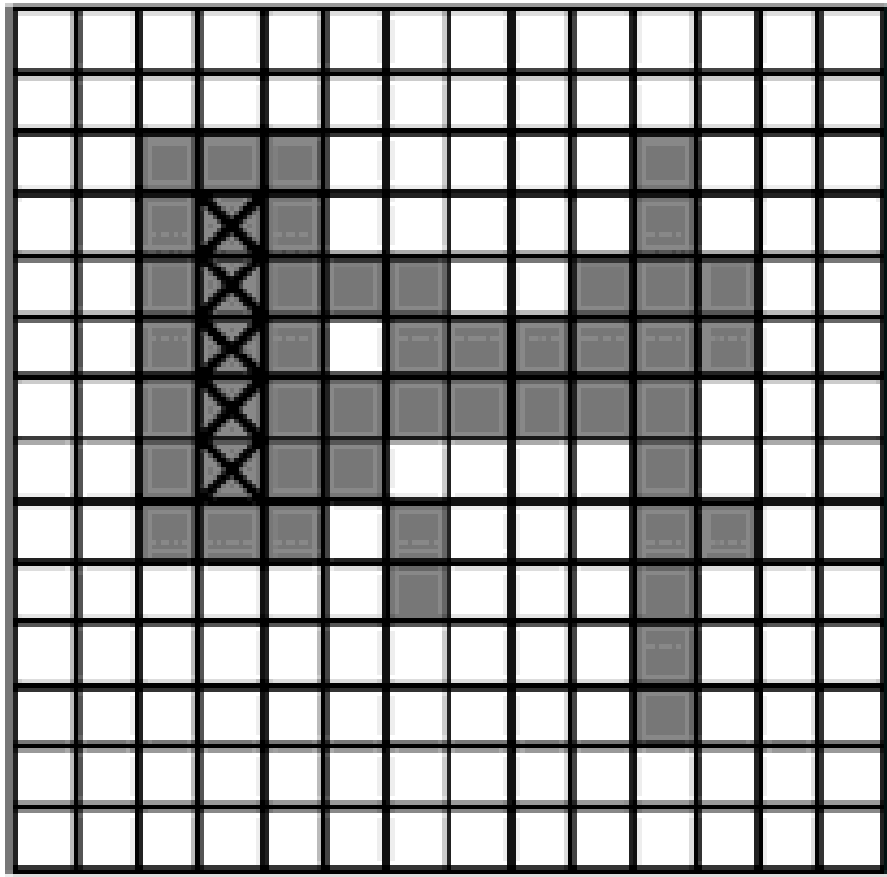


Processed Image

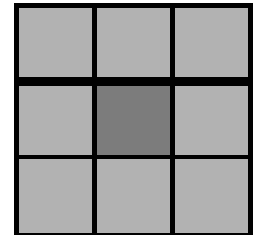


Structuring Element

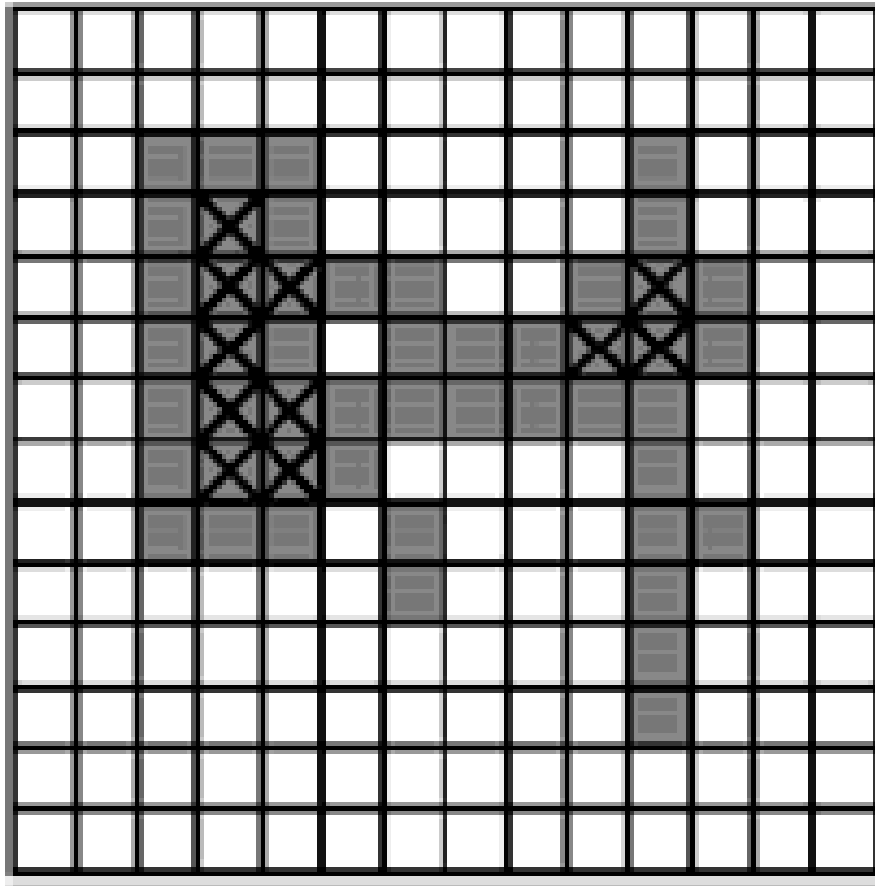
Erosion



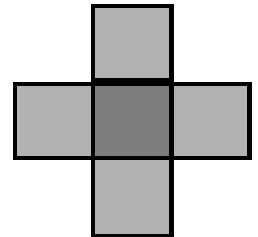
SE=



Erosion

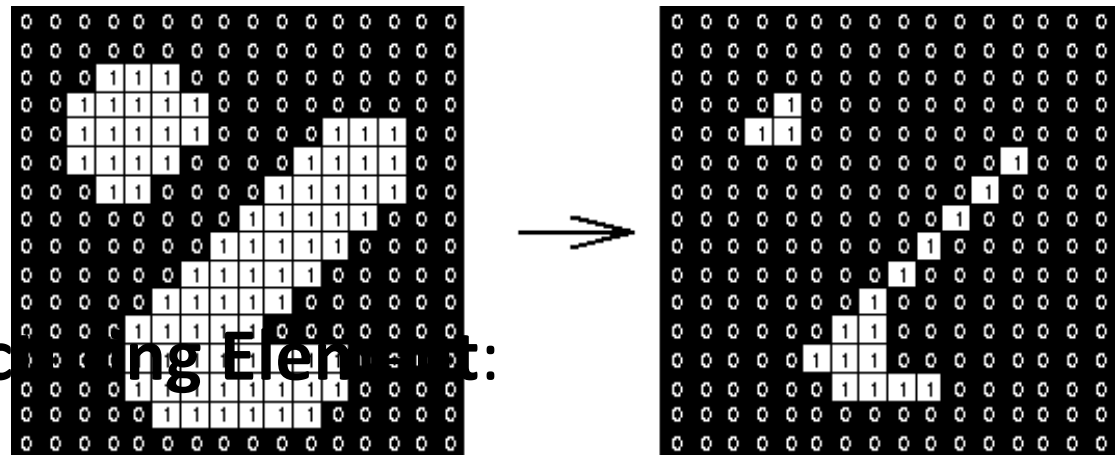


SE=



A first Example: Erosion

- **Erosion** is an important morphological operation



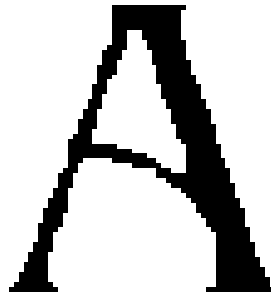
- Applied Structuring Element:

1	1	1
1	1	1
1	1	1

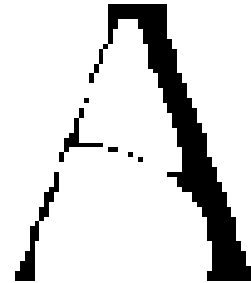
Erosion Example 1



Original image

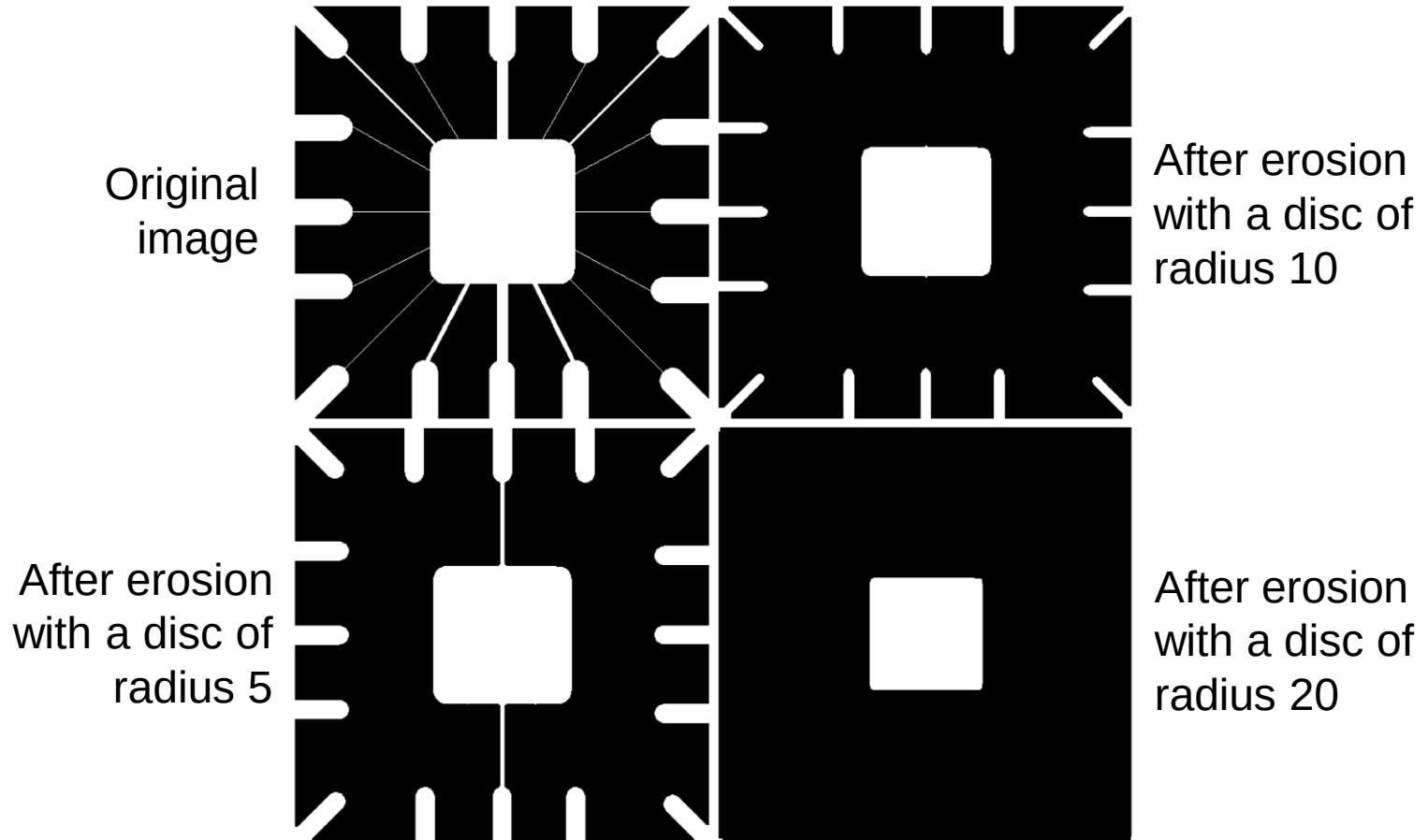


Erosion by 3*3
square structuring
element

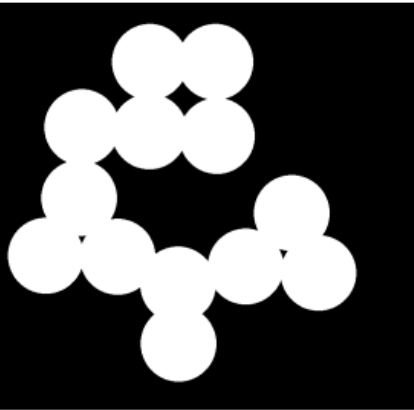


Erosion by 5*5
square structuring
element

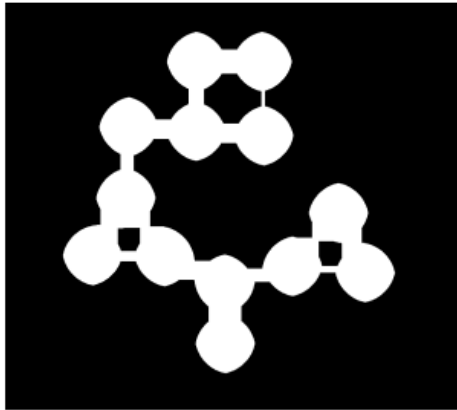
Erosion Example 2



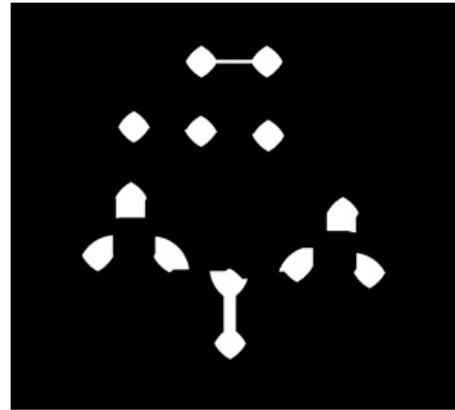
Erosion Example 3



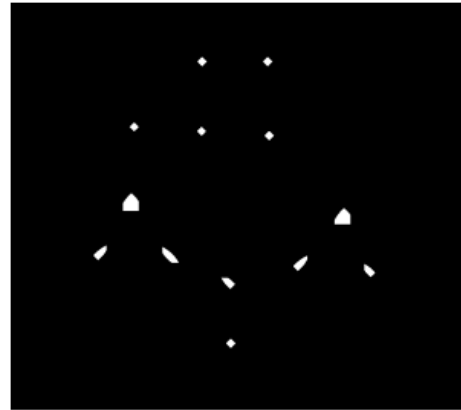
Original binary image
Circles (792x892)



Erosion by 30x30
structuring element



Erosion by 70x70
structuring element



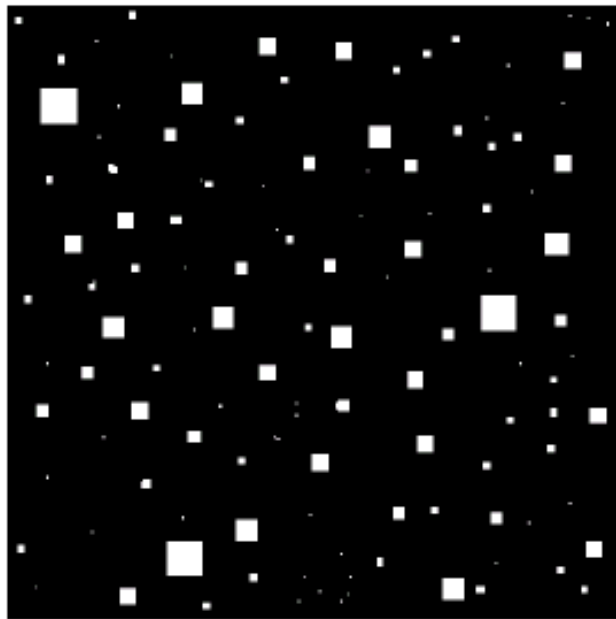
Erosion by 96x96
structuring element

Application of erosion

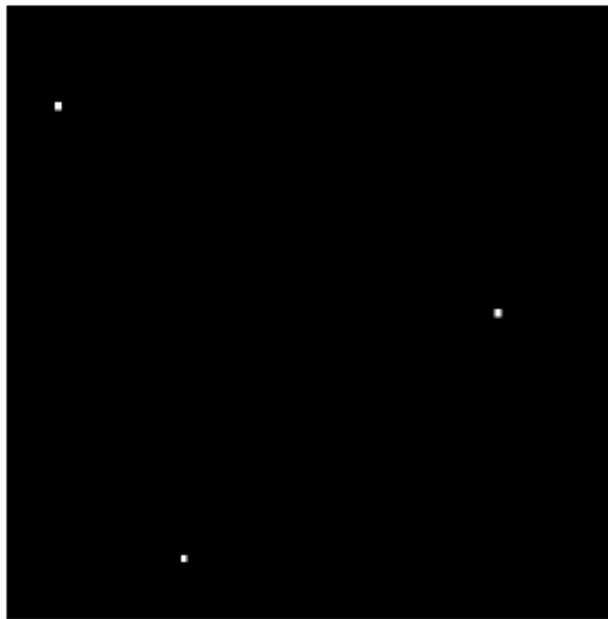
Loại bỏ các chi tiết không liên quan

Squares of size
1,3,5,7,9,15 pels

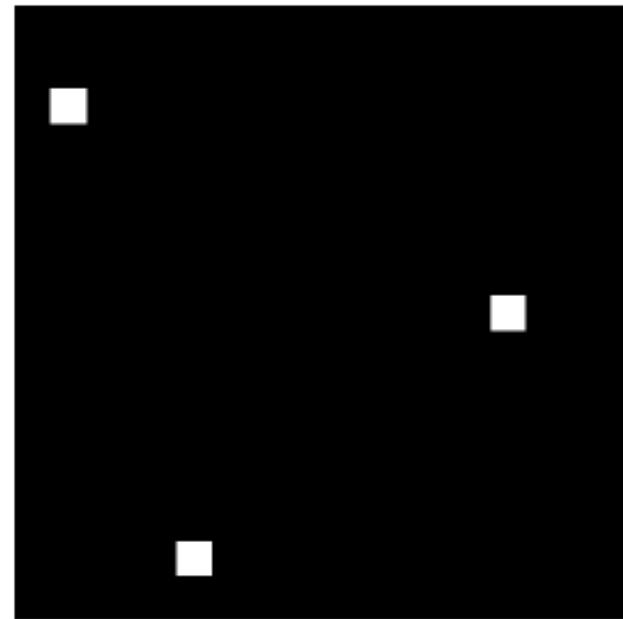
Erode with
13x13 square



original image



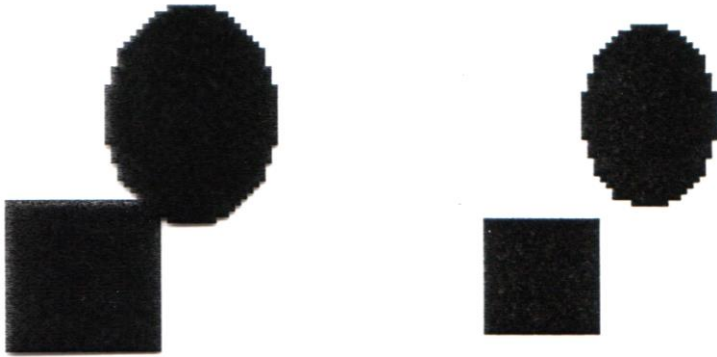
erosion



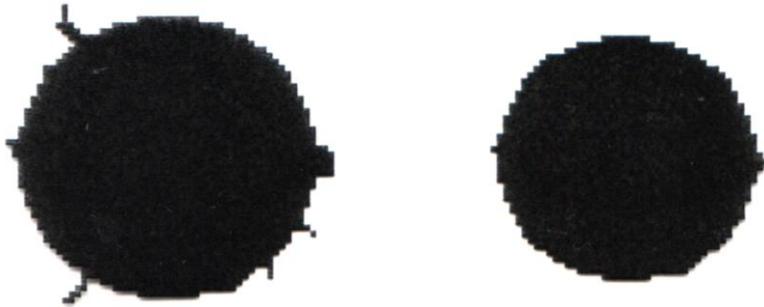
dilation

What Is Erosion For?

Erosion can split apart joined objects



Erosion can strip away extrusions



Dilation – Phép giãn

- Phép giãn ảnh f với phần tử cấu trúc s , ký hiệu:

$$f \oplus s$$

- Mô tả:

Giả sử SE s đang ở vị trí (x,y) , giá trị của pixel mới sau khi thực hiện phép toán có giá trị như sau:

$$g(x,y) = \begin{cases} 1 & \text{if } s \text{ hits } f \\ 0 & \text{otherwise} \end{cases}$$

Dilation – Phép giãn

Công thức:

$$\mathbf{A} \oplus \mathbf{B} = \{\mathbf{c} \mid \mathbf{c} = \mathbf{a} + \mathbf{b}, \mathbf{a} \in \mathbf{A}, \mathbf{b} \in \mathbf{B}\}$$

Trong đó:

- $\mathbf{A}$: Ma trận điểm ảnh của ảnh nhị phân.
- $\mathbf{B}$: Là phần tử cấu trúc.

Phép giãn nở (Dilation) ảnh sẽ cho ra một tập điểm ảnh $\mathbf{c}$ thuộc $\mathbf{D(i)}$, thấy rằng đây là một phép tổng giữa $\mathbf{A}$ và $\mathbf{B}$.

$\mathbf{A}$ sẽ là tập con của $\mathbf{D(i)}$.

Some Basic Definitions

- **Dilation – giãn/mở rộng**

$$A \oplus B = \{x \mid (B + x) \cap A \neq \emptyset\}$$

- Dilation expands a region.



Original (701x781)



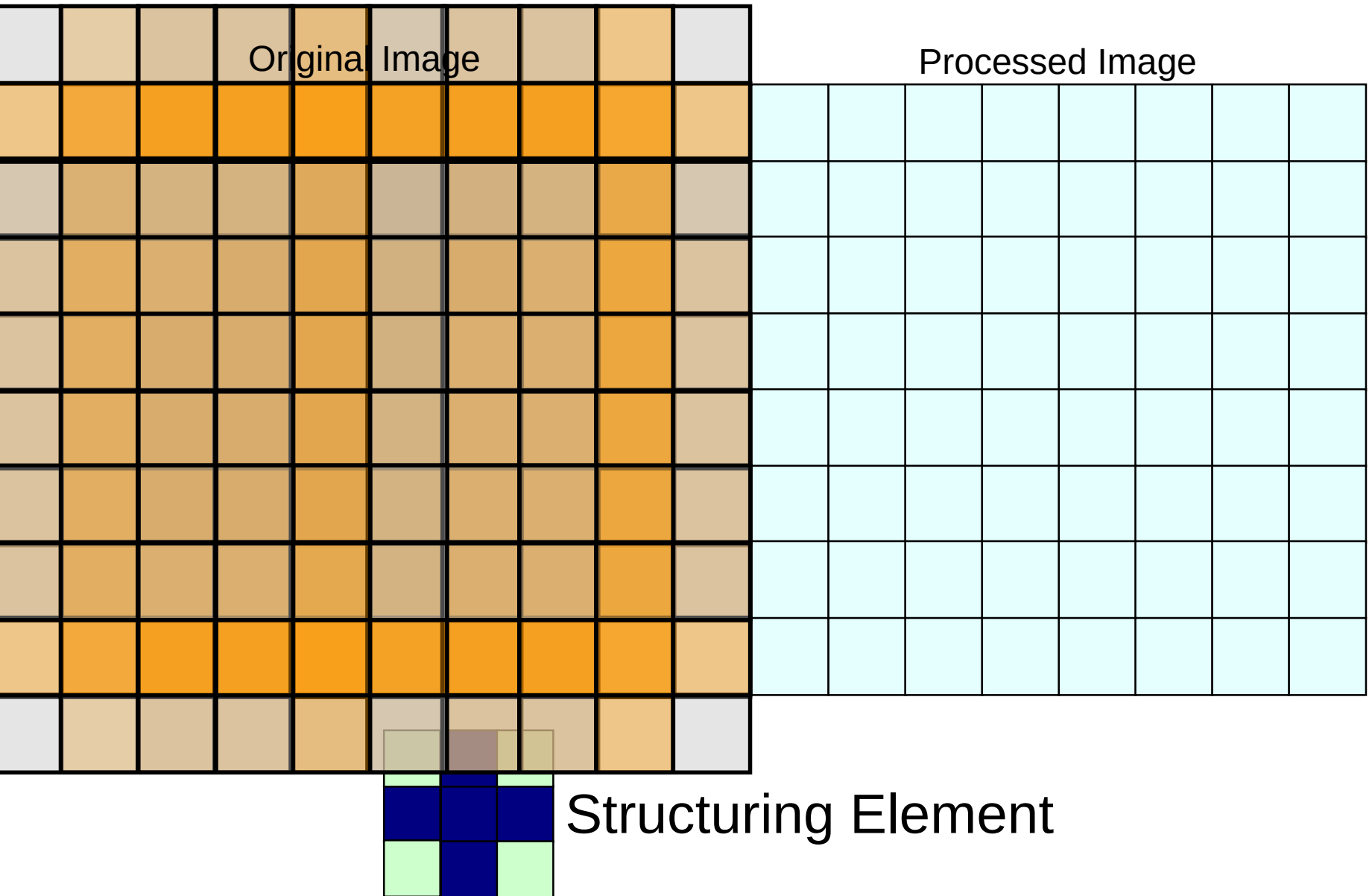
dilation with
3x3 structuring element



dilation with
7x7 structuring element

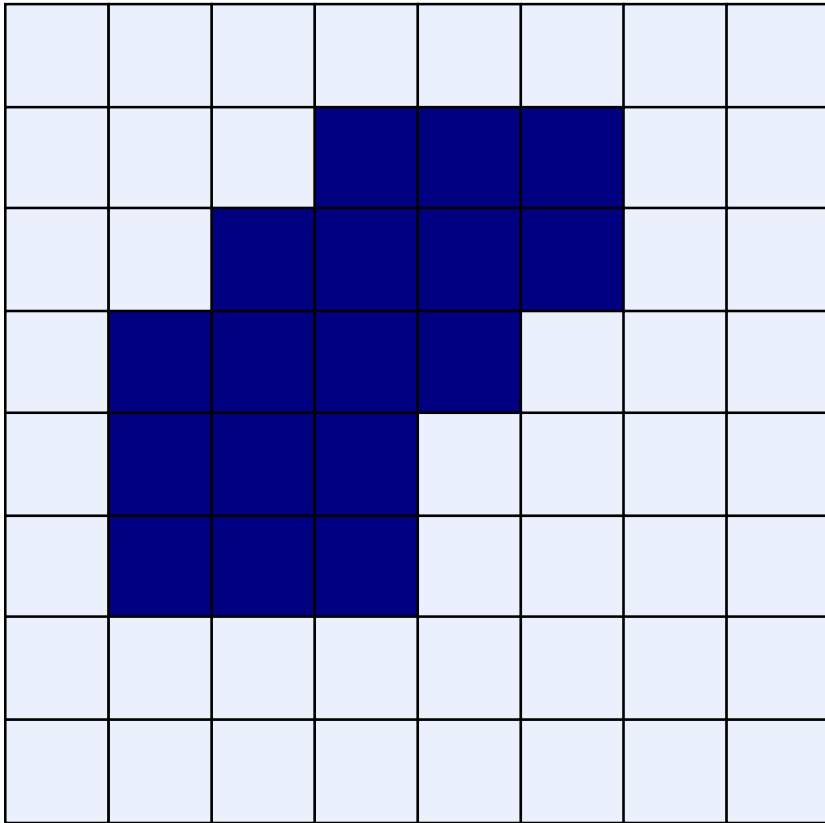
- Expands the size of 1-valued objects
- Smooths object boundaries
- Closes holes and gaps

Dilation Example

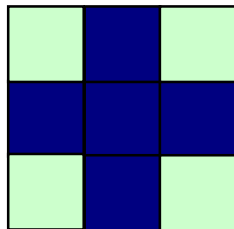
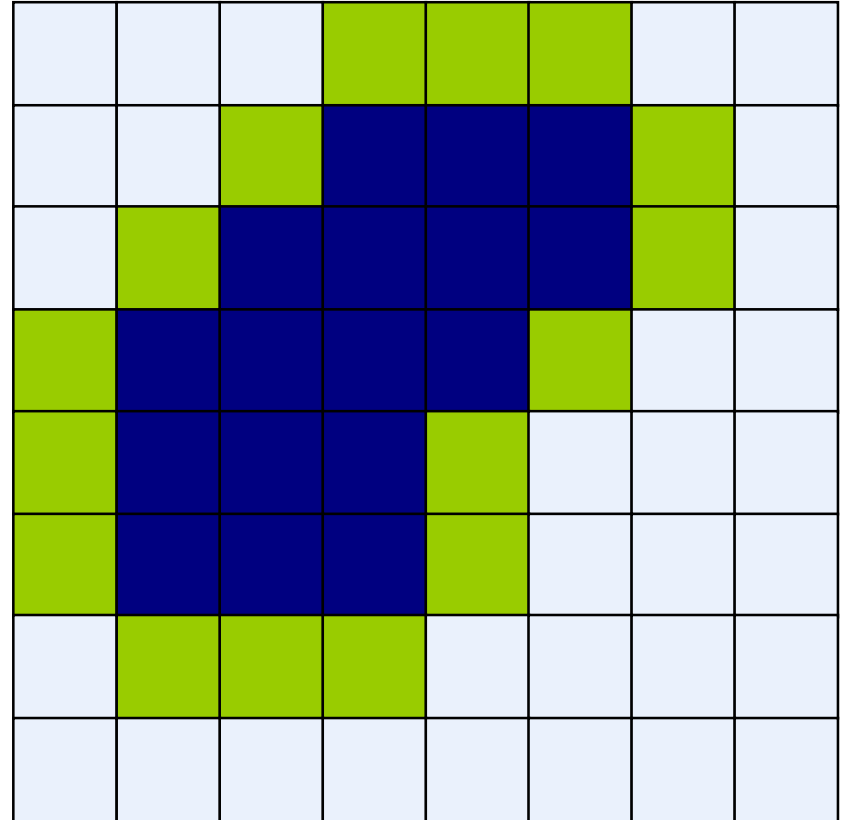


Dilation Example

Original Image

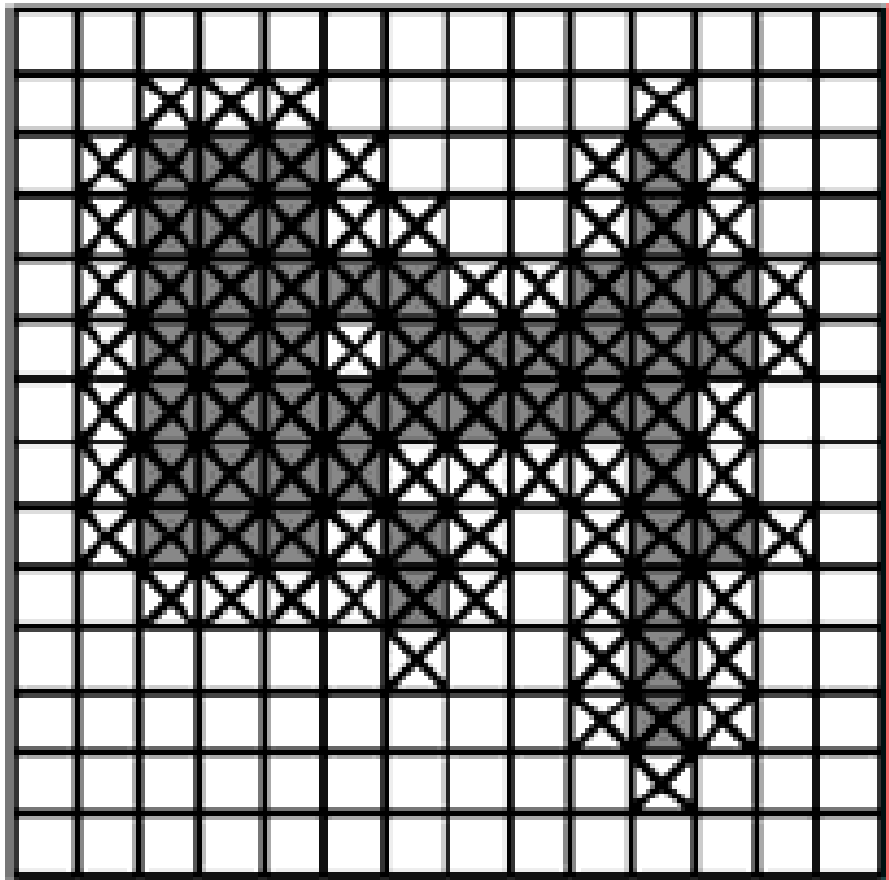


Processed Image With Dilated Pixels

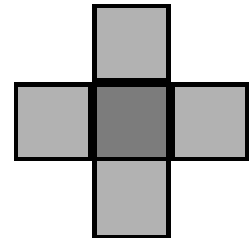


Structuring Element

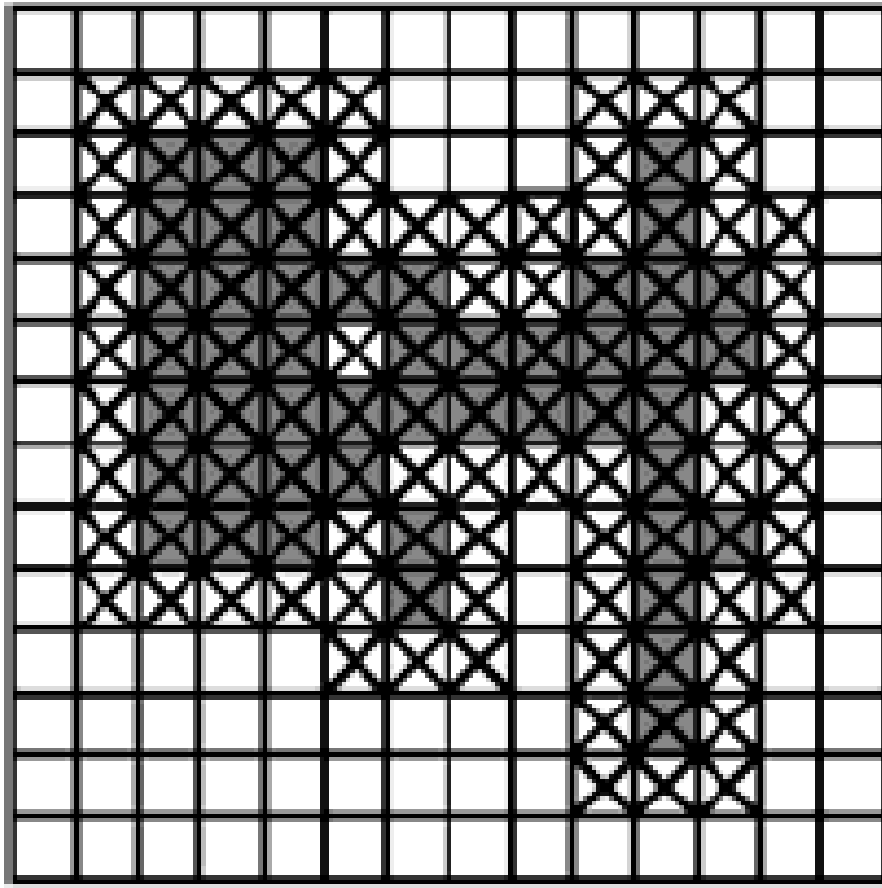
Dilation



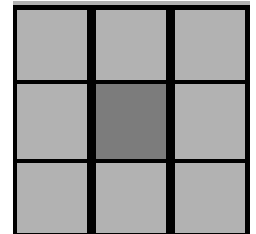
SE=



Dilation

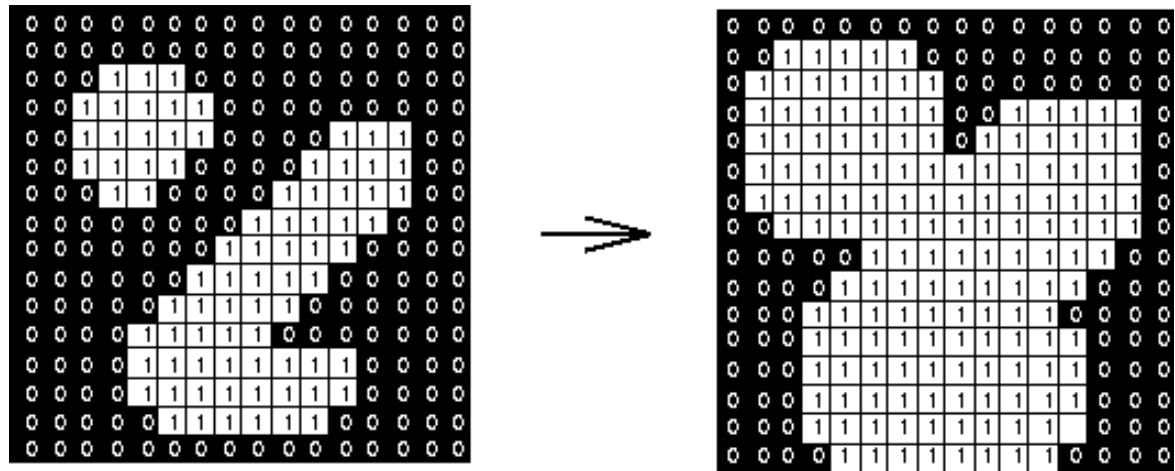


SE=



Example: Dilation

- **Dilation** is an important morphological operation



- **Applied Structuring Element:**

1	1	1
1	1	1
1	1	1

Dilation Example 1



Original image



Dilation by 3*3
square structuring
element



Dilation by 5*5
square structuring
element

Application of dilation: bridging gaps in images

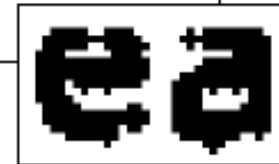
Original image

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



After dilation

Historically, certain computer programs were written using only two digits rather than four to define the applicable year. Accordingly, the company's software may recognize a date using "00" as 1900 rather than the year 2000.



0	1	0
1	1	1
0	1	0

Structuring element

What Is Dilation For?

Dilation can repair breaks

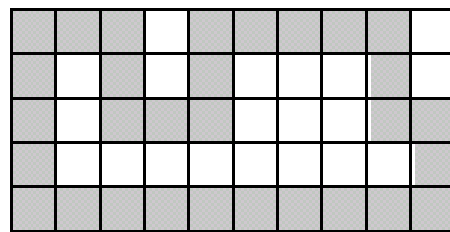
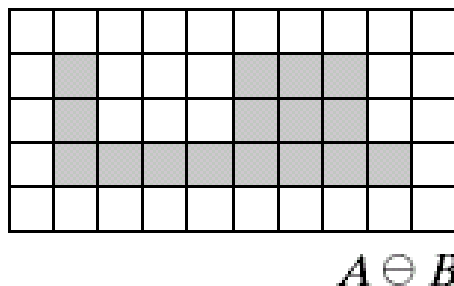
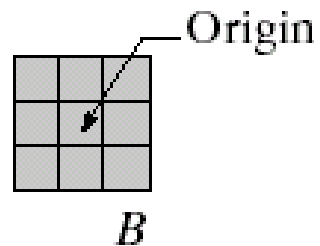
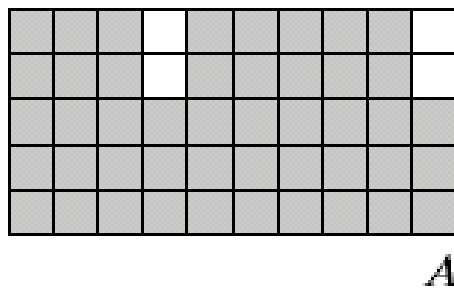


Dilation can repair intrusions



Dilation and erosion are duals

- Extract boundary of a set A :
 - First erode A (make A smaller)



Some Basic Definitions

- **Opening** is erosion followed by dilation:

$$A \circ B = (A \ominus B) \oplus B$$

- Opening smooths regions, removes spurs, breaks narrow lines.

- **Closing** is dilation followed by erosion:

$$A \bullet B = (A \oplus B) \ominus B$$

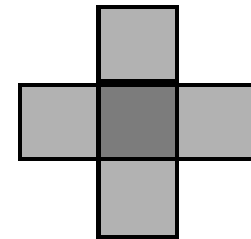
- Closing fills narrow gaps and holes in a region.

Bài tập

Bài 1: Cho ảnh A:

0	0	0	0	0	0	0	0	0
0	0	1	1	0	1	1	0	0
0	0	1	0	1	0	0	1	0
0	0	0	0	0	0	1	0	0
0	0	1	0	0	0	1	0	0
0	1	1	1	1	1	1	0	0
0	0	0	0	0	0	0	0	0

SE=



Thực hiện:

- a. Dilation
- b. Erosion

Bài tập

Bài 2: Cho ảnh X và phần tử cấu trúc B :

$$X = \begin{bmatrix} 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 1 & 1 & 0 \\ 0 & 1 & 1 & 0 & 1 & 1 & 1 & 1 \\ 1 & 1 & 0 & 1 & 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 & 0 & 1 & 1 & 1 \\ 1 & 0 & 1 & 1 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 & 1 & 1 & 1 \end{bmatrix} \quad B = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 1 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Thực hiện:

- Dilation
- Erosion

Some Basic Definitions

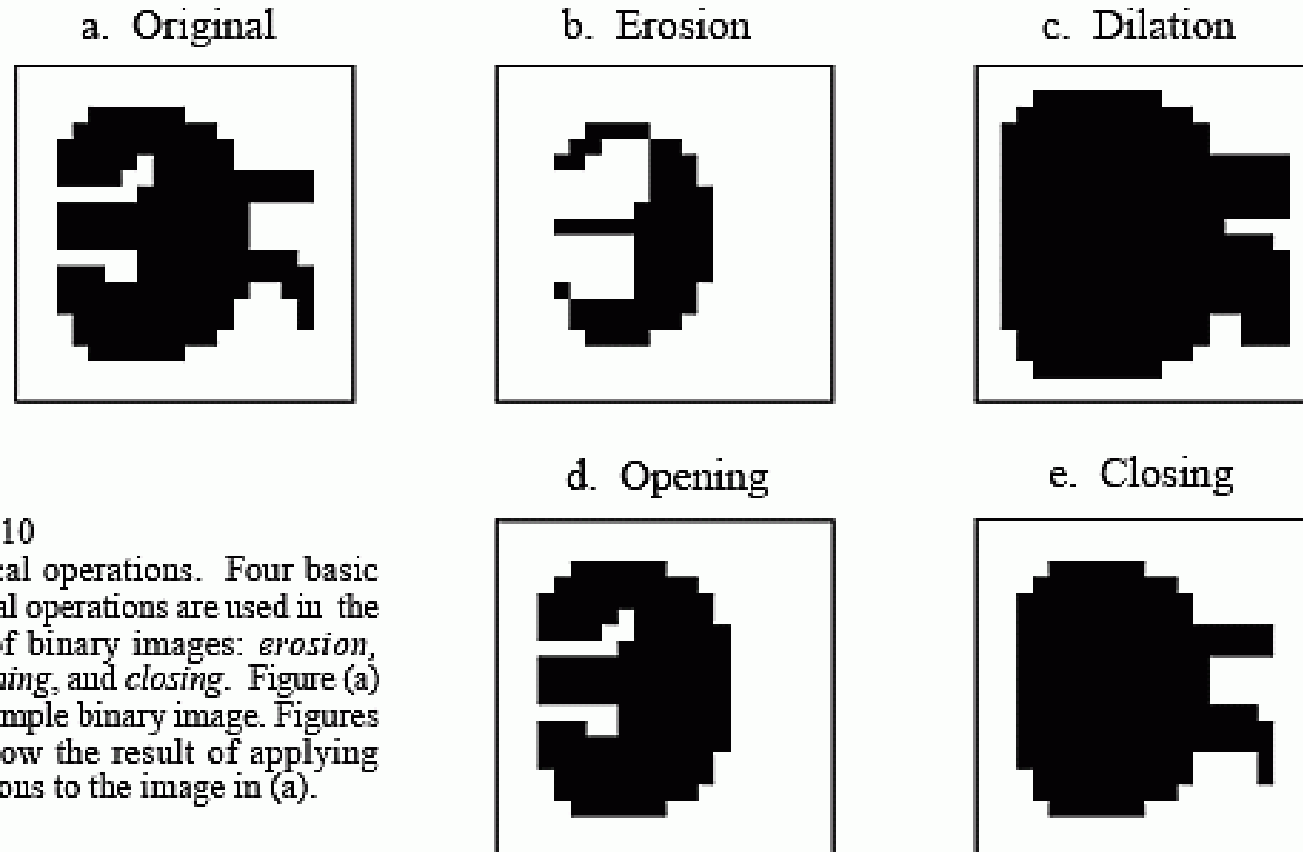
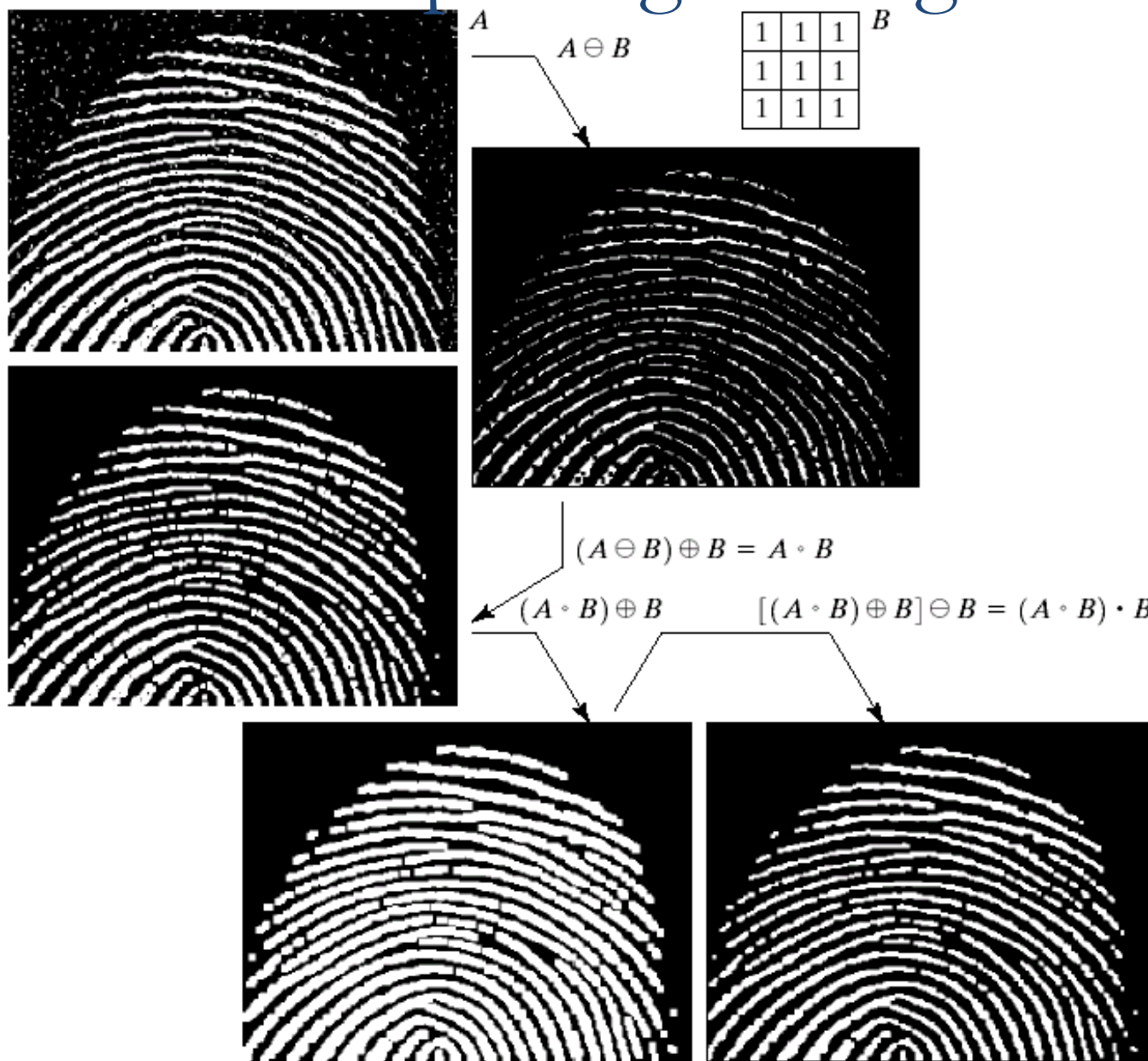


FIGURE 25-10
Morphological operations. Four basic morphological operations are used in the processing of binary images: *erosion*, *dilation*, *opening*, and *closing*. Figure (a) shows an example binary image. Figures (b) to (e) show the result of applying these operations to the image in (a).

Some Morphological Algorithms



a	b
d	c
e	f

FIGURE 9.11

(a) Noisy image.
 (c) Eroded image.
 (d) Opening of A .
 (d) Dilation of the opening.
 (e) Closing of the opening. (Original image for this example courtesy of the National Institute of Standards and Technology.)

Some Morphological Algorithms

- Boundary of a set, A, can be found by
 $A - (A \ominus B)$



a b

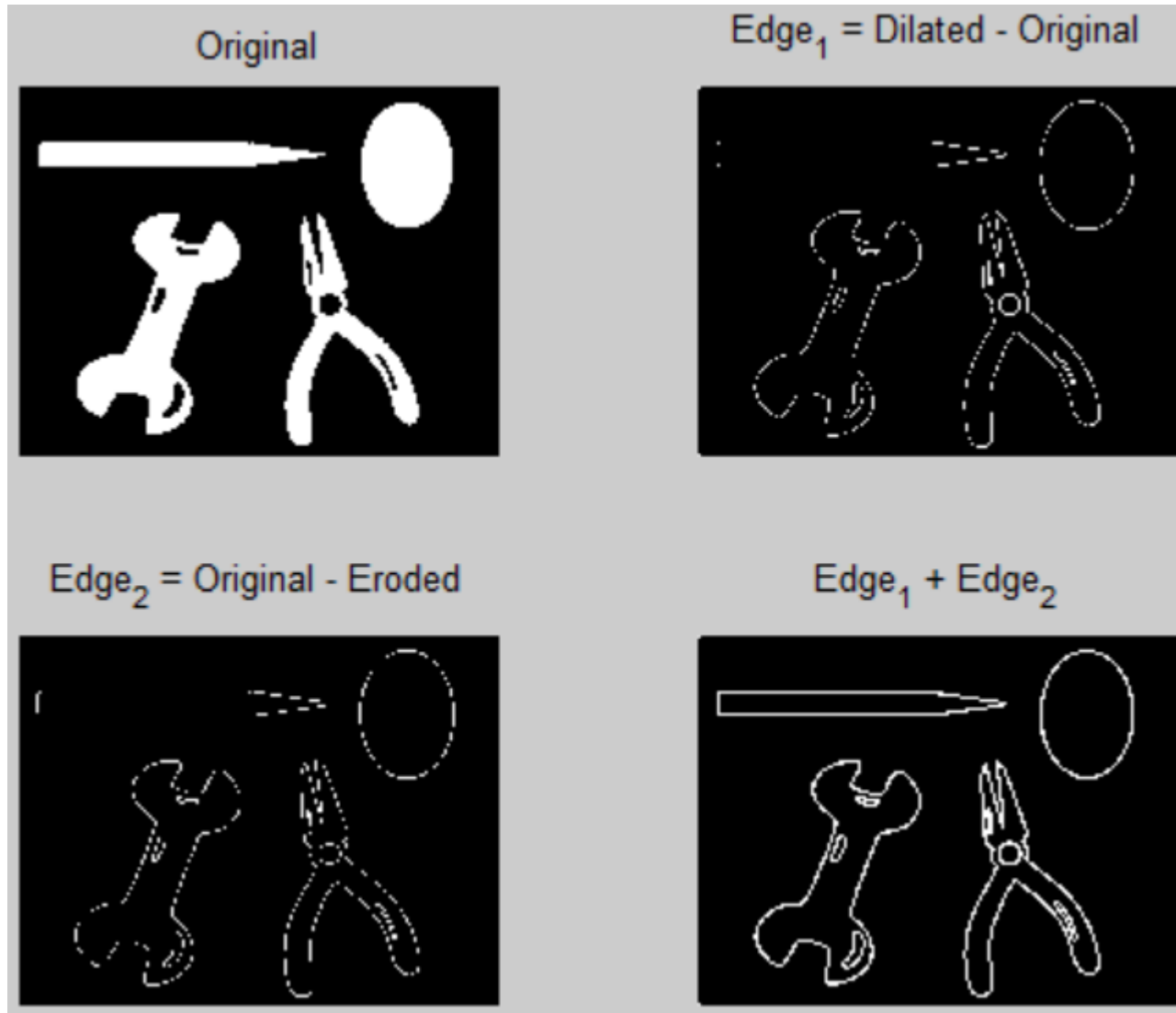
FIGURE 9.14

(a) A simple binary image, with 1's represented in white. (b) Result of using Eq. (9.5-1) with the structuring element in Fig. 9.13(b).

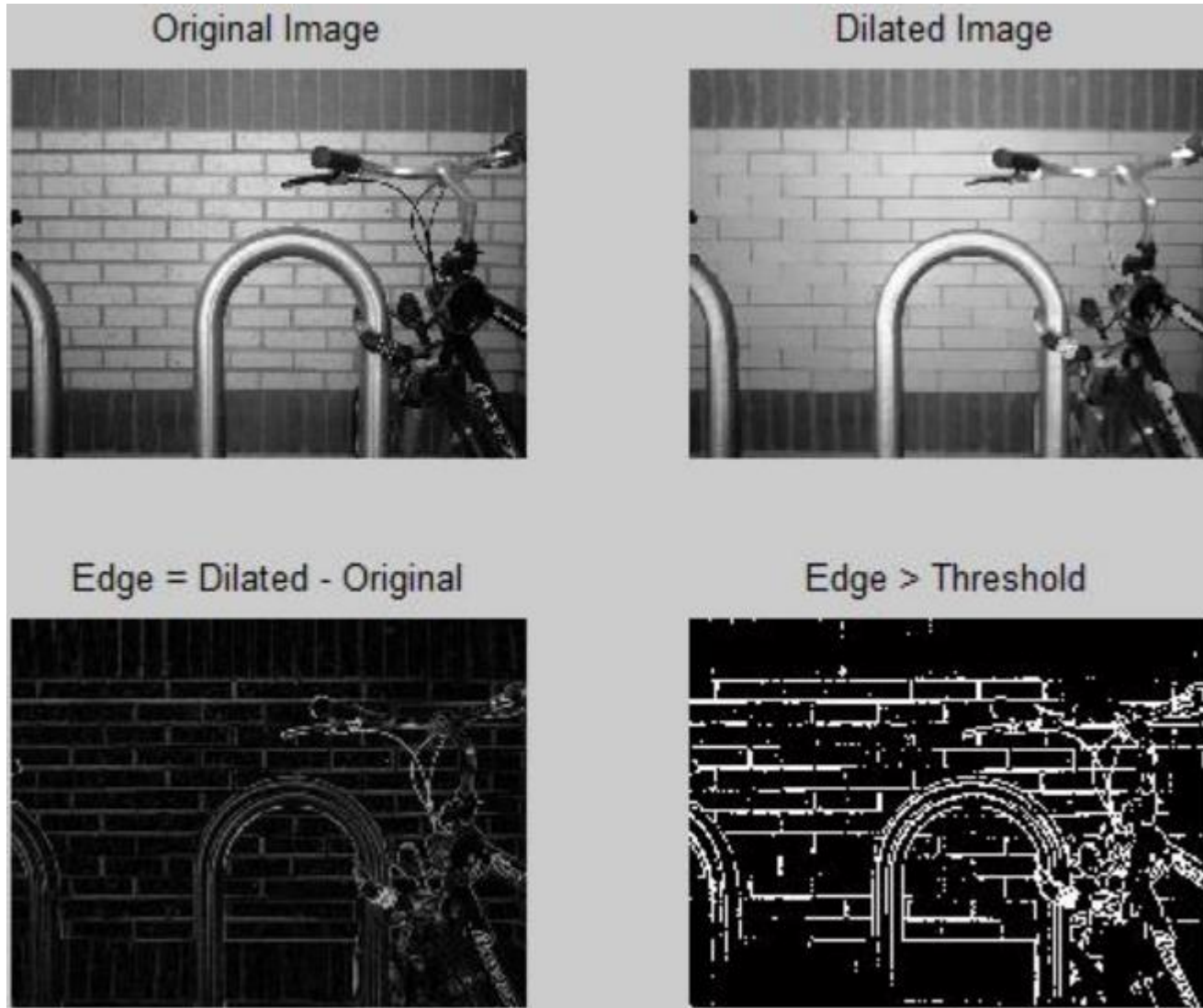


B

Some Morphological Algorithms



Some Morphological Algorithms



Summary

TABLE 9.2
Summary of
morphological
operations and
their properties.

		Comments (The Roman numerals refer to the structuring elements shown in Fig. 9.26).
Operation	Equation	
Translation	$(A)_z = \{w w = a + z, \text{ for } a \in A\}$	Translates the origin of A to point z .
Reflection	$\hat{B} = \{w w = -b, \text{ for } b \in B\}$	Reflects all elements of B about the origin of this set.
Complement	$A^c = \{w w \notin A\}$	Set of points not in A .
Difference	$A - B = \{w w \in A, w \notin B\}$ $= A \cap B^c$	Set of points that belong to A but not to B .
Dilation	$A \oplus B = \{z (\hat{B})_z \cap A \neq \emptyset\}$	“Expands” the boundary of A . (I)
Erosion	$A \ominus B = \{z (B)_z \subseteq A\}$	“Contracts” the boundary of A . (I)
Opening	$A \circ B = (A \ominus B) \oplus B$	Smooths contours, breaks narrow isthmuses, and eliminates small islands and sharp peaks. (I)
Closing	$A \bullet B = (A \oplus B) \ominus B$	Smooths contours, fuses narrow breaks and long thin gulfs, and eliminates small holes. (I)